GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Data Sheet for Piping Valves**
Document Number	EA-BFPCL-GPMC-GGPL-FEED-MPP-DSH-003
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Originator	Chinedu Anthony
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Data Sheet for Piping Valves

R01	05.07.22	Issued For Review	C. Anthony	E. Nwachukwu	A. Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R01	05/07/2022	Issue for Clients Review
R02		
A01		

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-MPP-DSH-003

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

TABLE OF CONTENTS

1. SCOPE

This document covers data sheets of Manual Ball Valves within BBOU manifold Station, Elepa Manifold Station and Akabel Pigging Station. excluding pipeline valves on the mainlines.

This document shall be read in conjunction with the project reference documents listed bellow.

1.1 PROJECT REFERENCE DOCUMENTS

S/No	Document Number	Description
1	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Class Specification
2	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003	Specifications for Piping Valves
3	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003	Process Engineering Flow Sheets 1 - 8
4	EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-010	Valve List.

1.2 CODES AND STANDARDS

S/No	Document Number	Description
1	API 607	Fire Test for Quarter-turn Valves and Valves Equipped with Nonmetallic Seats
2	ASME B16.5	Pipe Flanges and Flanged Fittings NPS ½ Through NPS 24
3	ASME B16.34	Valves – Flanged, Threaded, and Welding End
4	ASME B1.20.1	Pipe Threads, General Purpose (Inch)
5	BS EN ISO 6755-2	Testing of Valves - Part 2: Specification for Fire Type-Testing Requirements
6	EEMUA Publication No.182	Specification for integral block and bleed valve manifolds for direct connection to pipework
7	ISO 5208	Industrial valves - Pressure testing of metallic valves
8	ISO 14313	Petroleum and natural gas industries - Pipeline transportation systems - Pipeline Valves

3. ABBREVIATIONS

Abbreviation	Definition
AED	Anti-Explosive Decompression
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
CA	Corrosion Allowance
CS	Carbon Steel
DPT	Dye Penetrant Test
EN	European Standards
ENP	Electro-less Nickel Plating
FB	Full Bore
GR	Grade
HT	Hardness Test
LCB	Low Temperature (Wrought) Carbon Grade B
LTCS	Low Temperature Carbon Steel
MPI	Magnetic Particle Inspection
MSS	Manufacturers Standardization Society
NDT	Non-Destructive Testing
NPT-F	National Pipe Taper - Female
OS&Y	Outside Screw & Yolk
P-T	Pressure-Temperature
PEEK	Polyetheretherketone
PS	Pigging Station
RB	Reduced Bore
RF	Raised Face
RTJ	Ring Type Joint
RPTFE	Reinforced Polytetrafluoroethylene
RT	Radiographic Testing
SPWD	Spiral Wound
SS	Stainless Steel
SW	Socket Welded
TGS	Terminal Gas Station
THRD	Threaded
UT	Ultrasonic Testing
WCB	Wrought Carbon Grade B

5. IMPORTANT NOTES

5.1 This data sheet shall be completely filled by the vendor, stamped, signed and deviations clearly marked for acceptance. Quotation without this requirement will not be considered for technical review and evaluation.

5.2 Gland packing design shall take into account the requirement for low emission, low maintenance service. Graphite packing shall be low to medium density, high purity incorporating a passivating corrosion inhibitor and anti-extrusion elements.

5.3 Valves shall have position indicator showing open and close positions.

5.4 Minimum thickness of weld overlay shall be 3.0mm undiluted thickness as per trim material in final machined condition.

5.5 The wall thickness of valve body and other pressure containing parts shall include corrosion allowance specified in the datasheet in addition to the minimum wall thickness required as per ASME B16.34

5.6 Hydrostatic test to be performed on body and seat, low pressure pneumatic seat test to be performed on seat.

5.7 Machined surfaces shall be subject to 100% MPI/DPT.

5.8 Valve stem shall be of sufficient length to accommodate a minimum of 100mm insulation thickness If Isulation is specified on the data sheet.Valve stem shall be anti-blow out stem design.

5.9 For 2" & above trunnion mounted ball valves, provision shall be made for drain. For 8" & above vent and drain connection shall be foreseen.Vent and drain connections shall be as per ASME B16.34 / MSS SP-45 and with double O' rings, unless specified otherwise.

5.10 Lifting eyes / lug shall be provided for all valves of sizes 8" NPS and larger valves weighing 250 Kg and above, whichever is stringent, to facilitate maintenance. Lifting lugs shall be independent of body bolting.

5.11 Vendor shall confirm suitability of seat and seal material based on service fluids and service design conditions. Vendor shall submit P-T rating chart of selected seats /seals complying with the above at the time of bid as well as post order. Requirement listed in the data sheet & specification shall be considered as minimum.

5.12 For the given temperature in the data sheet, valve seat insert (soft) material shall be able to withstand the limiting pressure indicated in rating table of ASME B16.34. Reduction in limiting pressure of valve due to limitation of seat insert material is not acceptable.

5.13 Valve bolting material shall be able to withstand the pressure as indicated in full rating pressure table of ASME B16.34. Vendor shall submit bolt strength calculation report for the same for review.

5.14 All soft seated SW ball valves shall supply with pup piece of 100 mm on both side of valve. Pup piece shall be welded by the valve manufacturer and under its responsibility prior to valve testing. 2 nos. of test rings shall be supplied by manufacturer per material and size.

5.15 All elastomeric materials (o-rings, seals, seats, etc.) shall be resistant to explosive decompression (ED) and resistant to absorption of hydrocarbon liquids.

5.16 Valve stems shall be one-piece design and made from bar / forged material.

5.17 Valves shall be tested and qualified for both vertical & horizontal positions as per applicable standards indicated in datasheet. Change of field installation position from the valve test position shall not be considered as a cause for valve leakage if any.

5.18 Stem & seat sealant injection connection shall be suitable for injecting sealant / grease with the valve in operating condition.

5.19 All sealant injection connections shall incorporate an internal non-return valve, a giant button head and a cover with vent holes that seals off the fitting by plugging the sealant port.

5.20 Valves shall not be shipped or tested with sealant injected into the valves.

5.21 For flanged-end valves, body and flanges shall be integrally forged.

5.22 Galvanised Bolts and Nuts used in valves shall be hot dip Per ASTM A153.

5.23 All threaded end valves used for instrument connections shall be NPT (Female) threaded to ASME B1.20.1.

5.24 All manual valves (whether operated with a lever or a gear operator) shall be provided with locking device plate to facilitate heavy duty padlocks or car-seals to prevent operation in the locked open and close position.

5.25 Color of Top Coat shall be Aluminium / Light Grey (RAL 9006 / 7035). Flange faces should not be painted. The Colour shall be discusses with Vendor before PO.

5.26 The use of asbestos material in any form is not permitted.

5.27 Valves shall have a double block and bleed feature, in the open and closed position to facilitate flushing, draining and venting of the valve.

5.28 Valve support shall be provided for valves weighing 250 kg or more; valve support material shall be same as that of valve body.

5.29 Vendor shall provide Inspection and Testing Plan based on the specifications and standards mentioned.

5.30 "O" ring groove dimensions and surface finish shall be in accordance with BS 4518. " O" rings shall be Anti Explosive Decompression certified for Gas service and compatible with methanol.

S/N	Datasheet 1: Datasheet for Ball Valve, Carbon Steel, SW with Pup Piece, CL 800, 0.5" to 1.5", FB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 1.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	0.5" to 1.5"		
6	Valve ASME Class / Rating	800#		
7	Valve Port	Full Bore		
8	Valve Service	Hydrocarbon Gas		
9	Valve End Connection	Socket Welded with 100mm Pup Piece (As Per Table1.1 Below)		
10	Corrosion Allowance	As Per Table 1.1 Below		
11	Design Temperature (Min / Max)	**- 29 / 150 ^{o}C**		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Hand Lever		
15	Body Construction	Split Body, Long Pattern		
16	Ball Support	Floating Ball		
17	Lifting Lugs	Not required		
18	Sealant Injection Facility	Not required		
19	Vent / Drain Requirements	Not required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
1	05.07.22	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid) / Trunnion	ASTM A182 316L		
29	Stem / Trim	ASTM A182 316L		
30	Pup Piece Material / Length	ASTM A106 Gr. B / 100mm (Note 5.14)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	SPWD; SS304 Winding Graphite Filled; SS304 Inner Ring; CS Outer Ring		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	Test Ring	Required (Note 5.14)		
41	**Codes and Standards / Testing / Certification**			
42	Valve Design Code	As Per ASME B16.34 / API 608 / ISO 17292		
43	Face to Face Dimensions	As Per ASME B16.10 / Manufacturer's Standard		
44	Fire Safe	As Per API 607		
45	Hydrostatic & Pneumatic Test	As Per API 598		
46	Low Pressure Gas Seat Test	As Per API 598 at 6 barg		
47	Leakage Test	Leak Rate "A" As Per ISO 5208		
48 49	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
50 51	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
52 53 54	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
55 56	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
57 58	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

59	Table 1.1 (As Per Valve List)									
60 61	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
62	1	BBOU Manifold Station	B03	3.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1010	BBOU-6"-B-003-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
63	2	BBOU Manifold Station	B03	3.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1015	BBOU-6"-B-004-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
64	3	BBOU Manifold Station	B03	3.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1050	BBOU-6"-B-006-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
65	4	BBOU Manifold Station	B03	3.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1048	BBOU-6"-B-007-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
66	5	EPEPA Manifold Station	B06	6.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1010	ELEP-6"-B-003-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
67	6	EPEPA Manifold Station	B06	6.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1015	ELEP-6"-B-004-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
68	7	EPEPA Manifold Station	B06	6.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1066	ELEP-6"-B-005-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
69	8	EPEPA Manifold Station	B06	6.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1050	ELEP-6"-B-006-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
70	9	EPEPA Manifold Station	B06	6.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1048	ELEP-6"-B-007-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
71	10	Akabel Pigging Station	B06	6.0mm	3/4''	800#	Socket Welded with 100mm Pup	MV1066	AKABP-6"-B-005-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 2: Datasheet for Ball Valve, Carbon Steel, SW with Pup Piece, CL 1500, 0.5" to 1.5", FB		
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**
2	**General Data**		
3	Piping Class	As Per Table 2.1 Below	
4	Valve Type	Ball Valve	
5	Valve Size Range	0.5" to 1.5"	
6	Valve ASME Class / Rating	1500#	
7	Valve Port	Full Bore	
8	Valve Service	Hydrocarbon Gas	
9	Valve End Connection	Socket Welded with 100mm Pup Piece (As Per Table 2.1 Below)	
10	Corrosion Allowance	As Per Table 2.1 Below	
11	Design Temperature (Min / Max)	- 29 / 150 °C	
12	Design Pressure	Full rating as per ASME B16.34	
13	**Design**		
14	Valve Operation	Hand Lever	
15	Body Construction	Split Body, Long Pattern	
16	Ball Support	Floating Ball	
17	Lifting Lugs	Not required	
18	Sealant Injection Facility	Not required	
19	Vent / Drain Requirements	Not required	
20	Locking Facility	Required	
21	Stem	Anti-Blow-Out Type	
22	Seat	Self-Relieving	
23	Cavity Pressure Relief	Self-Relieving Seat	
24	Anti-Static Design	Required	
25	**Materials**		
1	05.07.22	ASTM A105N	
27	Cover / Bonnet	ASTM A105N	
28	Ball (Solid) / Trunnion	ASTM A182 316L	
29	Stem / Trim	ASTM A182 316L	
30	Pup Piece Material / Length	ASTM A106 Gr. B / 100mm (Note 5.14)	
31	Seat Insert	PEEK (Note 5.11)	
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)	
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)	
34	Anti-Static Device / Spring	INCONEL 718	
35 36	Gaskets	SPWD; SS304 Winding Graphite Filled; SS304 Inner Ring; CS Outer Ring	
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) *(Note 5.22)*	
39	Hand Lever	Malleable Steel / Nodular Iron	
40	Test Ring	Required (Note 5.14)	
41	**Codes and Standards / Testing / Certification**		
42	Valve Design Code	As Per ASME B16.34 / API 608 / ISO 17292	
43	Face to Face Dimensions	As Per ASME B16.10 / Manufacturer's Standard	
44	Fire Safe	As Per API 607	
45	Hydrostatic & Pneumatic Test	As Per API 598	
46	Low Pressure Gas Seat Test	As Per API 598 at 6 barg	
47	Leakage Test	Leak Rate "A" As Per ISO 5208	
48 49	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
50 51	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
52 53 54	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)	
55 56	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
57 58	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)	

59	Table 2.1 (As Per Valve List)									
60 61	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
62	1	BBOU Manifold Station	C03	3.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1004	BBOU-4"-P-004-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
63	2	BBOU Manifold Station	C03	3.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1008	BBOU-6"-P-006-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
64	3	BBOU Manifold Station	C03	3.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1013	BBOU-6"-P-007-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
65	4	BBOU Manifold Station	CO3	3.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1046	BBOU-6"-P-011-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
66	5	BBOU Manifold Station	CO3	3.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1026	BBOU-24"-P-008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
67	6	EPEPA Manifold Station	C06	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1004	ELEP-4"-P-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
68	7	EPEPA Manifold Station	C06	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1008	ELEP-6"-P-006-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
69	8	EPEPA Manifold Station	C06	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1013	ELEP-6"-P-007-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
70	9	EPEPA Manifold Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1064	ELEP-4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
71	10	EPEPA Manifold Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1061	ELEP-8"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
72	11	EPEPA Manifold Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1060	ELEP-24"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
73	12	EPEPA Manifold Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1046	ELEP-6"-P-021-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
74	13	EPEPA Manifold Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1026	ELEP-36"-P-017-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
75	15	Akabel Pigging Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1064	AKABP -4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
76	16	Akabel Pigging Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1061	AKABP-12"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
77	17	Akabel Pigging Station	CO6	6.0mm	3/4''	1500#	Socket Welded with 100mm Pup Piece	MV1060	AKABP-36"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 3: Datasheet for Ball Valve, Carbon Steel, Flanged RF, CL 150, 2" to 4", RB			
1	Description	Specification Requirement	Vendor's Confirmation	
2	General Data			
3	Piping Class	As Per Table 3.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	2" to 4"		
6	Valve ASME Class / Rating	150#		
7	Valve Port	Reduced Bore		
8	Valve Service	Hydrocarbon Gas / Open / Close Drain		
9	Valve End Connection	Flanged RF with smooth finish (Ra 3.2 to 6.4 µm)		
10	Corrosion Allowance	As Per Table 3.1 Below		
11	Design Temperature (Min / Max)	- 29 / 150 °C		
12	Design Pressure	Full rating as per ASME B16.34		
13	Design			
14	Valve Operation	Hand Lever		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Floating Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Not required		
19	Vent / Drain Requirements	Not required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	Materials			
1	05.07.22	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	Spiral Wound, Low Stress, ASME B16.20, Ss 316 Winding 4.5mm Thick, Graphite Filler, Inner and Centering Ring 3.2 mm Thick		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) *(Note 5.22)*		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	Codes and Standards / Testing / Certification			
41	Valve Design Code	As Per API 6D / ISO 14313		
42	Face to Face Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

60	Table 3.1 (As Per Valve List)									
61 62	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
63	1	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED	MV1054	BBOU-4"-D-008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
64	2	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED	E2R	BBOU-6"-B-001-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
65	3	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED	E2R	BBOU-4"-D-001-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
66	4	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED		NOZZLE 3 BBOU-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
67	5	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED		NOZZLE 4 BBOU-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
68	6	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED		NOZZLE 5 BBOU-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
69	7	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED	E2R	BBOU-6"-B-007-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
70	8	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED		INST. NOZZLE BBOU-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
71	9	BBOU Manifold Station	B03	3.0mm	2"	150#	RF FLANGED		INST. NOZZLE BBOU-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03
72	10	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED	E2R	ELEP-6"-B-001-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
73	11	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED	E2R	BBOU-4"-D-001-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
74	12	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED		NOZZLE 3 ELEP-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
75	13	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED		NOZZLE 4 ELEP--CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
76	14	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED		NOZZLE 5 ELEP--CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
77	15	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED	E2R	BBOU-6"-B-007-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
78	16	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED		INST. NOZZLE ELEP-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
79	17	EPEPA Manifold Station	B06	6.0mm	2"	150#	RF FLANGED		INST. NOZZLE ELEP-CDT-001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_07
80	18	EPEPA Manifold Station	B06	6.0mm	4"	150#	RF FLANGED	MV1054	ELEP-4"-D-011-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
81	19	Akabel Pigging Station	B06	6.0mm	2"	150#	RF FLANGED	MV1076	AKABP-2"-D-008-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 4: Datasheet for Ball Valve, Carbon Steel, Flanged RTJ, CL 900, 2", RB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 4.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	2"		
6	Valve ASME Class / Rating	900#		
7	Valve Port	Reduced Bore		
8	Valve Service	Hydrocarbon Gas / Open / Close Drain		
9	Valve End Connection	Flanged RTJ		
10	Corrosion Allowance	As Per Table 4.1 Below		
11	Design Temperature (Min / Max)	- 29 / 150 °C		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Hand Lever		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Trunnion Mounted Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Not required		
19	Vent / Drain Requirements	Not required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
1	05.07.22	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	**Codes and Standards / Testing / Certification**			
41	Valve Design Code	As Per API 6D / ISO 14313		
42	Face to Face Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

60	Table 4.1 (As Per Valve List)									
61 62	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
63	1	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1002	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
64	2	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1018	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
65	3	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1019	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
66	4	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1020	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
67	5	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1047	BBOU-6"-P-011-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
68	6	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1029	PIG TRAP.BBOU-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
69	7	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1030	PIG TRAP.BBOU-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
70	8	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1025	BBOU-8"-P-009-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
71	9	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1024	BBOU-8"-P-009-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
72	10	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1021	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
73	11	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1028	BBOU-24"-P-008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
74	12	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1035	PIG TRAP.BBOU-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
75	13	BBOU Manifold Station	C03	3.0mm	2"	900#	RTJ FLANGED	MV1036	PIG TRAP.BBOU-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
76	14	Elepa Manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	MV1018	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
77	15	Elepa Manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	MV1019	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
78	16	Elepa Manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	MV1020	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
79	17	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1073	PIG TRAP. ELEP-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
80	18	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1063	ELEP-8"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
81	19	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1069	PIG TRAP. ELEP-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
82	20	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1068	PIG TRAP. ELEP-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
83	21	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1058	ELEP-24"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
84	22	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1047	ELEP-6"-P-021-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
85	23	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1029	PIG TRAP. ELEP-A-300	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
86	24	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1030	PIG TRAP. ELEP-A-300	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
87	25	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1025	ELEP-12"-P-019-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
88	26	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1024	ELEP-12"-P-019-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
89	27	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1021	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
90	28	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1028	ELEP-36"-P-017-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
91	29	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1035	PIG TRAP. ELEP-A-300	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
92	30	Elepa Manifold Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1036	PIG TRAP. ELEP-A-300	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
93	31	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1073	PIG TRAP. AKAB-A-100	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
94	32	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1063	AKABP-12"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
95	33	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1069	PIG TRAP. AKAB-A-100	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
96	34	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1068	PIG TRAP. AKAB-A-100	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
97	35	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	MV1058	AKABP-36"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 5: Datasheet for Ball Valve, Carbon Steel, Flanged RTJ, CL 1500, 2", RB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 5.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	2"		
6	Valve ASME Class / Rating	1500#		
7	Valve Port	Reduced Bore		
8	Valve Service	Hydrocarbon Gas / Open/ Close Drain		
9	Valve End Connection	Flanged RTJ		
10	Corrosion Allowance	As Per Table 5.1 Below		
11	Design Temperature (Min / Max)	- 29 / 150 °C		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Hand Lever		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Trunnion Mounted Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Not required		
19	Vent / Drain Requirements	Not required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
1	05.07.22	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	**Codes and Standards / Testing / Certification**			
41	Valve Design Code	As Per API 6D / ISO 14313		
42	Face to Face Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		
60	**Table 5.1 (As Per Valve List)**			

	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
61 62	**S/N**	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
63	1	Elepa Manifold Station	D06	6.0mm	2"	1500#	RTJ FLANGED	MV1002	ELEP-36"-P-002-D06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
64										

S/N	Datasheet 6: Datasheet for Ball Valve, Duplex Stainless Steel , Flanged RTJ, CL 900, 2", RB		
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**
2	**General Data**		
3	Piping Class	As Per Table 6.1 Below	
4	Valve Type	Ball Valve	
5	Valve Size Range	2"	
6	Valve ASME Class / Rating	900#	
7	Valve Port	Reduced Bore	
8	Valve Service	Hydrocarbon Gas (Corrosive)	
9	Valve End Connection	Flanged RTJ	
10	Corrosion Allowance	As Per Table 6.1 Below	
11	Design Temperature (Min / Max)	- 29 / 150 °C	
12	Design Pressure	Full rating as per ASME B16.34	
13	**Design**		
14	Valve Operation	Hand Lever	
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature	
16	Ball Support	Trunnion Mounted Ball	
17	Lifting Lugs	Required for valves weighing 250kg or heavier	
18	Sealant Injection Facility	Not required	
19	Vent / Drain Requirements	Not required	
20	Locking Facility	Required	
21	Stem	Anti-Blow-Out Type	
22	Seat	Self-Relieving	
23	Cavity Pressure Relief	Self-Relieving Seat	
24	Anti-Static Design	Required	
25	**Materials**		
1	05.07.22	ASTM A182 F51	
27	Cover / Bonnet	ASTM A182 F51	
28	Ball (Solid)	UNS S31803	
29	Stem / Trim	UNS S31803	
30	Seat Housing	UNS S31803	
31	Seat Insert	PEEK (Note 5.11)	
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)	
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)	
34	Anti-Static Device / Spring	INCONEL 718	
35	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, TYPE 316	
36			
37	Bolting	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	
38			
39	Hand Lever	Malleable Steel / Nodular Iron	
40	**Codes and Standards / Testing / Certification**		
41	Valve Design Code	As Per API 6D / ISO 14313	
42	Face to Face Dimensions	As Per API 6D	
43	End Connection	As Per ASME B16.5	
44	Fire Safe	As Per API 6FA / BS EN ISO 10497	
45	Hydrostatic & Pneumatic Test	As Per API 6D	
46	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)	
47			
48	Leakage Test	Leak Rate "A" As Per ISO 5208	
49	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
50			
51	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
52			
53	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)	
54			
55			
56	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
57			
58	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)	
59			
60	**Table 6.1 (As Per Valve List)**		

	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
61 / 62										
63	1	BBOU Manifold Station	C70	0.0mm	2"	900#	RTJ FLANGED	MV1001	BBOU-24"-P-001-C70	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
64	2									

S/N	Datasheet 7: Datasheet for Ball Valve, Duplex Stainless Steel , Flanged RTJ, CL 1500, 2", RB		
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**
2	**General Data**		
3	Piping Class	As Per Table 7.1 Below	
4	Valve Type	Ball Valve	
5	Valve Size Range	2"	
6	Valve ASME Class / Rating	1500#	
7	Valve Port	Reduced Bore	
8	Valve Service	Hydrocarbon Gas (Corrosive)	
9	Valve End Connection	Flanged RTJ	
10	Corrosion Allowance	As Per Table 7.1 Below	
11	Design Temperature (Min / Max)	**- 29 / 150 ^{o}C**	
12	Design Pressure	Full rating as per ASME B16.34	
13	**Design**		
14	Valve Operation	Hand Lever	
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature	
16	Ball Support	Trunnion Mounted Ball	
17	Lifting Lugs	Required for valves weighing 250kg or heavier	
18	Sealant Injection Facility	Not required	
19	Vent / Drain Requirements	Not required	
20	Locking Facility	Required	
21	Stem	Anti-Blow-Out Type	
22	Seat	Self-Relieving	
23	Cavity Pressure Relief	Self-Relieving Seat	
24	Anti-Static Design	Required	
25	**Materials**		
1	05.07.22	ASTM A182 F51	
27	Cover / Bonnet	ASTM A182 F51	
28	Ball (Solid)	UNS S31803	
29	Stem / Trim	UNS S31803	
30	Seat Housing	UNS S31803	
31	Seat Insert	PEEK (Note 5.11)	
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)	
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)	
34	Anti-Static Device / Spring	INCONEL 718	
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, TYPE 316	
37 38	Bolting	STUD BOLTS, 2 HEAVY HEX NUTS, ASTM A453 GRADE 660 CLASS B	
39	Hand Lever	Malleable Steel / Nodular Iron	
40	**Codes and Standards / Testing / Certification**		
41	Valve Design Code	As Per API 6D / ISO 14313	
42	Face to Face Dimensions	As Per API 6D	
43	End Connection	As Per ASME B16.5	
44	Fire Safe	As Per API 6FA / BS EN ISO 10497	
45	Hydrostatic & Pneumatic Test	As Per API 6D	
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)	
48	Leakage Test	Leak Rate "A" As Per ISO 5208	
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)	
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)	
60	**Table 7.1 (As Per Valve List)**		

	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
61 62	**S/N**	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
63	1	Elepa Manifold Station	D70	0.0mm	2"	1500#	RTJ FLANGED	MV1001	ELEP-36"-P-001-D70	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
64	2									

S/N	Datasheet 8: Datasheet for Ball Valve, Carbon Steel, Flanged RF, CL 150, 6", RB		
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**
2	**General Data**		
3	Piping Class	As Per Table 8.1 Below	
4	Valve Type	Ball Valve	
5	Valve Size Range	6"	
6	Valve ASME Class / Rating	150#	
7	Valve Port	Full Bore	
8	Valve Service	Hydrocarbon Gas	
9	Valve End Connection	Flanged RF with smooth finish (Ra 3.2 to 6.4 μm)	
10	Corrosion Allowance	As Per Table 8.1 Below	
11	Design Temperature (Min / Max)	- 29 / 150 °C	
12	Design Pressure	Full rating as per ASME B16.34	
13	**Design**		
14	Valve Operation	Hand Lever	
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature	
16	Ball Support	Trunnion Mounted Ball	
17	Lifting Lugs	Required for valves weighing 250kg or heavier	
18	Sealant Injection Facility	Not required	
19	Vent / Drain Requirements	Not required	
20	Locking Facility	Required	
21	Stem	Anti-Blow-Out Type	
22	Seat	Self-Relieving	
23	Cavity Pressure Relief	Self-Relieving Seat	
24	Anti-Static Design	Required	
25	**Materials**		
1	05.07.22	ASTM A105N	
27	Cover / Bonnet	ASTM A105N	
28	Ball (Solid)	A105N + ENP	
29	Stem / Trim	A105N + ENP	
30	Seat Housing	ASTM A182 F6A (13CR)	
31	Seat Insert	PEEK (Note 5.11)	
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)	
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)	
34	Anti-Static Device / Spring	INCONEL 718	
35 36	Gaskets	Spiral Wound, Low Stress, ASME B16.20, Ss 316 Winding 4.5mm Thick, Graphite Filler, Inner and Centering Ring 3.2 mm Thick	
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) *(Note 5.22)*	
39	Hand Lever	Malleable Steel / Nodular Iron	
40	**Codes and Standards / Testing / Certification**		
41	Valve Design Code	As Per API 6D / ISO 14313	
42	Face to Face Dimensions	As Per API 6D	
43	End Connection	As Per ASME B16.5	
44	Fire Safe	As Per API 6FA / BS EN ISO 10497	
45	Hydrostatic & Pneumatic Test	As Per API 6D	
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)	
48	Leakage Test	Leak Rate "A" As Per ISO 5208	
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)	
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)	
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)	

60	Table 8.1 (As Per Valve List)									
61 62	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
63	1	BBOU Manifold Station	B03	3.0mm	6"	150#	RF FLANGED	MV1051	BBOU-6"-B-001-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
64	2	BBOU Manifold Station	B03	3.0mm	6"	150#	RF FLANGED	MV1011	BBOU-6"-B-003-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
65	3	BBOU Manifold Station	B03	3.0mm	6"	150#	RF FLANGED	MV1016	BBOU-6"-B-004-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
66	4	BBOU Manifold Station	B03	3.0mm	6"	150#	RF FLANGED	MV1052	BBOU-6"-B-006-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
67	5	BBOU Manifold Station	B03	3.0mm	6"	150#	RF FLANGED	MV1049	BBOU-6"-B-007-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
68	6	EPEPA Manifold Station	B06	6.0mm	6"	150#	RF FLANGED	MV1051	ELEP-6"-B-001-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
69	7	EPEPA Manifold Station	B06	6.0mm	6"	150#	RF FLANGED	MV1011	ELEP-6"-B-003-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
70	8	EPEPA Manifold Station	B06	6.0mm	6"	150#	RF FLANGED	MV1016	ELEP-6"-B-004-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
71	9	EPEPA Manifold Station	B06	6.0mm	6"	150#	RF FLANGED	MV1051	ELEP-6"-B-005-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
72	10	EPEPA Manifold Station	B06	6.0mm	6"	150#	RF FLANGED	MV1067	ELEP-6"-B-005-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
73	11	EPEPA Manifold Station	B06	6.0mm	6"	150#	RF FLANGED	MV1052	ELEP-6"-B-006-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
74	12	EPEPA Manifold Station	B06	6.0mm	6"	150#	RF FLANGED	MV1049	ELEP-6"-B-007-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
75	13									
76	14	Akabel Pigging Station	B06	6.0mm	6"	150#	RF FLANGED	MV1051	AKABP-6"-B-005-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
77	15	Akabel Pigging Station	B06	6.0mm	6"	150#	RF FLANGED	MV1067	AKABP-6"-B-005-B06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 9: Datasheet for Ball Valve, Carbon Steel, Flanged RTJ, CL 900, 4" and Above, RB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 9.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	4" and above		
6	Valve ASME Class / Rating	900#		
7	Valve Port	Reduced Bore		
8	Valve Service	Hydrocarbon Gas / Open/ Close Drain		
9	Valve End Connection	Flanged RTJ		
10	Corrosion Allowance	As Per Table 9.1 Below		
11	Design Temperature (Min / Max)	**- 29 / 150 °C**		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Gear Operated Hand Wheel		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Trunnion Mounted Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Required for Size 6" and Above		
19	Vent / Drain Requirements	Required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
1	05.07.22	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) *(Note 5.22)*		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	**Codes and Standards / Testing / Certification**			
41	Valve Design Code	As Per API 6D / ISO 14313		
42	Face to Face Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

60	Table 9.1 (As Per Valve List)									
61 62	S/N	Location	Pipe Class	Corrossion Allowance	Size	Rating	End Connection	Valve Tag	Line Number	Reference P&ID Number
63	1	BBOU Manifold Station	C03	3.0mm	24"	900#	RTJ FLANGED	SDV1001	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
64	2	BBOU Manifold Station	C03	3.0mm	24"	900#	RTJ FLANGED	MV1043	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
65	3	BBOU Manifold Station	C03	3.0mm	24"	900#	RTJ FLANGED	MV1044	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
66	4	BBOU Manifold Station	C03	3.0mm	24"	900#	RTJ FLANGED	MV1045	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
67	5	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	MV1003	BBOU-4"-P-004-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
68	6	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	MV1005	BBOU-4"-P-004-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
69	7	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	BDV1001	BBOU-4"-P-004-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
70	8	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	MV1006	BBOU-4"-P-004-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
71	9	BBOU Manifold Station	C03	3.0mm	6"	900#	RTJ FLANGED	MV1007	BBOU-6"-P-006-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
72	10	BBOU Manifold Station	C03	3.0mm	6"	900#	RTJ FLANGED	MV1009	BBOU-6"-P-006-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
73	11	BBOU Manifold Station	C03	3.0mm	6"	900#	RTJ FLANGED	MV1012	BBOU-6"-P-007-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
74	12	BBOU Manifold Station	C03	3.0mm	6"	900#	RTJ FLANGED	MV1014	BBOU-6"-P-007-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
75	13	BBOU Manifold Station	C03	3.0mm	24"	900#	RTJ FLANGED	MV1017	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
76	14	BBOU Manifold Station	C03	3.0mm	6"	900#	RTJ FLANGED	MV1031	BBOU-6"-P-011-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
77	15	BBOU Manifold Station	C03	3.0mm	8"	900#	RTJ FLANGED	MV1032	BBOU-8"-P-009-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
78	16	BBOU Manifold Station	C03	3.0mm	8"	900#	RTJ FLANGED	MV1023	BBOU-8"-P-009-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
79	17	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	MV1033	BBOU-4"-P-010-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
80	18	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	MV1037	BBOU-4"-D-006-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
81	19	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	MV1038	BBOU-4"-D-006-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
82	20	BBOU Manifold Station	C03	3.0mm	4"	900#	RTJ FLANGED	MV1053	BBOU-4"-D-006-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
83	21	Elepa Manifold Station	C06	6.0mm	36"	900#	RTJ FLANGED	MV1043	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
84	22	Elepa Manifold Station	C06	6.0mm	36"	900#	RTJ FLANGED	MV1044	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
85	23	Elepa Manifold Station	C06	6.0mm	36"	900#	RTJ FLANGED	MV1045	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
86	24	Elepa Manifold Station	C06	6.0mm	4"	900#	RTJ FLANGED	MV1003	ELEP-4"-P-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
87	25	Elepa Manifold Station	C06	6.0mm	4"	900#	RTJ FLANGED	MV1005	ELEP-4"-P-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
88	26	Elepa Manifold Station	C06	6.0mm	4"	900#	RTJ FLANGED	BDV1001	ELEP-4"-P-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
89	27	Elepa Manifold Station	C06	6.0mm	4"	900#	RTJ FLANGED	MV1006	ELEP-4"-P-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
90	28	Elepa Manifold Station	C06	6.0mm	6"	900#	RTJ FLANGED	MV1007	ELEP-6"-P-006-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
91	29	Elepa Manifold Station	C06	6.0mm	6"	900#	RTJ FLANGED	MV1009	ELEP-6"-P-006-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
92	30	Elepa Manifold Station	C06	6.0mm	6"	900#	RTJ FLANGED	MV1012	ELEP-6"-P-007-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
93	31	Elepa Manifold Station	C06	6.0mm	6"	900#	RTJ FLANGED	MV1014	ELEP-6"-P-007-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
94	32	Elepa Manifold Station	C06	6.0mm	36"	900#	RTJ FLANGED	MV1017	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
95	33	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1065	ELEP-4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
96	34	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1072	ELEP-4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
97	35	Elepa Manifold Station	CO6	6.0mm	8"	900#	RTJ FLANGED	MV1071	ELEP-8"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
98	36	Elepa Manifold Station	CO6	6.0mm	8"	900#	RTJ FLANGED	MV1062	ELEP-8"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
99	37	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1070	ELEP-4"-P-015-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	38	Elepa Manifold Station	CO6	6.0mm	24"	900#	RTJ FLANGED	MV1017	ELEP-24"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	39	Elepa Manifold Station	CO6	6.0mm	24"	900#	RTJ FLANGED	MV1059	ELEP-24"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	40	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1074	ELEP-4"-D-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	41	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1075	ELEP-4"-D-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	42	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1053	ELEP-4"-D-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	43	Elepa Manifold Station	CO6	6.0mm	6"	900#	RTJ FLANGED	MV1031	ELEP-6"-P-021-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
##	44	Elepa Manifold Station	CO6	6.0mm	12"	900#	RTJ FLANGED	MV1032	ELEP-12"-P-019-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
##	45	Elepa Manifold Station	CO6	6.0mm	12"	900#	RTJ FLANGED	MV1023	ELEP-12"-P-019-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
##	46	Elepa Manifold Station	CO6	6.0mm	24"	900#	RTJ FLANGED	MV1022	ELEP-36"-P-003-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
##	47	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1033	ELEP-4"-P-020-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
##	48	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1037	ELEP-4"-D-006-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
##	49	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1038	ELEP-4"-D-006-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
##	50	Elepa Manifold Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1053	ELEP-4"-D-006-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06

##	51	Akabel Pigging Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1065	AKABP -4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	52	Akabel Pigging Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1072	AKABP -4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	53	Akabel Pigging Station	CO6	6.0mm	12"	900#	RTJ FLANGED	MV1071	AKABP-12"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	54	Akabel Pigging Station	CO6	6.0mm	12"	900#	RTJ FLANGED	MV1062	AKABP-12"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	55	Akabel Pigging Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1070	AKABP-4"-P-015-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	56	Akabel Pigging Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1074	AKABP-4"-D-002-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
##	57	Akabel Pigging Station	CO6	6.0mm	4"	900#	RTJ FLANGED	MV1075	AKABP-4"-D-002-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 10: Datasheet for Ball Valve, Carbon Steel, Flanged RTJ, CL 900, 4" and Above, FB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 10.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	4" and above		
6	Valve ASME Class / Rating	900#		
7	Valve Port	Full Bore		
8	Valve Service	Hydrocarbon Gas / Open/ Close Drain		
9	Valve End Connection	Flanged RTJ		
10	Corrosion Allowance	As Per Table 10.1 Below		
11	Design Temperature (Min / Max)	**- 29 / 150 °C**		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Gear Operated Hand Wheel		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Trunnion Mounted Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Required for Size 6" and Above		
19	Vent / Drain Requirements	Required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
1	05.07.22	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	**Codes and Standards / Testing / Certification**			
41	Valve Design Code	As Per API 6D / ISO 14313		
42	Face to Face Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

60	**Table 10.1 (As Per Valve List)**									
61 62	**S/N**	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
63	1	BBOU Manifold Station	C03	3.0mm	24"	900#	RTJ FLANGED	MV1022	BBOU-24"-P-008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
64	2	BBOU Manifold Station	C03	3.0mm	24"	900#	RTJ FLANGED	MV1027	BBOU-24"-P-008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
65	3	Elepa Manifold Station	C06	6.0mm	36"	900#	RTJ FLANGED	MV1022	ELEP-36"-P-017-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
66	4	Elepa Manifold Station	C06	6.0mm	36"	900#	RTJ FLANGED	MV1027	ELEP-36"-P-017-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
67	5	Akabel Pigging Station	C06	6.0mm	36"	900#	RTJ FLANGED	MV1059	AKABP-36"-P-013-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 11: Datasheet for Ball Valve, Carbon Steel, Flanged RTJ, CL 1500, 4" and Above, FB			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	
2	**General Data**			
3	Piping Class	As Per Table 11.1 Below		
4	Valve Type	Ball Valve		
5	Valve Size Range	4" and above		
6	Valve ASME Class / Rating	1500#		
7	Valve Port	Ruduced Bore		
8	Valve Service	Hydrocarbon		
9	Valve End Connection	Flanged RTJ		
10	Corrosion Allowance	As Per Table 11.1 Below		
11	Design Temperature (Min / Max)	- 29 / 150 °C		
12	Design Pressure	Full rating as per ASME B16.34		
13	**Design**			
14	Valve Operation	Gear Operated Hand Wheel		
15	Body Construction	Split Body, Long Pattern with Double Block & Bleed Feature		
16	Ball Support	Trunnion Mounted Ball		
17	Lifting Lugs	Required for valves weighing 250kg or heavier		
18	Sealant Injection Facility	Required for Size 6" and Above		
19	Vent / Drain Requirements	Required		
20	Locking Facility	Required		
21	Stem	Anti-Blow-Out Type		
22	Seat	Self-Relieving		
23	Cavity Pressure Relief	Self-Relieving Seat		
24	Anti-Static Design	Required		
25	**Materials**			
1	05.07.22	ASTM A105N		
27	Cover / Bonnet	ASTM A105N		
28	Ball (Solid)	A105N + ENP		
29	Stem / Trim	A105N + ENP		
30	Seat Housing	ASTM A182 F6A (13CR)		
31	Seat Insert	PEEK (Note 5.11)		
32	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.30) (Note 5.11)		
33	Gland Packing	Filled Graphite (Note 5.2) (Note 5.11)		
34	Anti-Static Device / Spring	INCONEL 718		
35 36	Gaskets	RING JOINT, OCTAGONAL, CL900, ASME B16.20, SOFT IRON		
37 38	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized) ***(Note 5.22)***		
39	Hand Lever	Malleable Steel / Nodular Iron		
40	**Codes and Standards / Testing / Certification**			
41	Valve Design Code	As Per API 6D / ISO 14313		
42	Face to Face Dimensions	As Per API 6D		
43	End Connection	As Per ASME B16.5		
44	Fire Safe	As Per API 6FA / BS EN ISO 10497		
45	Hydrostatic & Pneumatic Test	As Per API 6D		
46 47	Low Pressure Gas Seat Test	Type II As Per API 6D (***at 6 barg*** with Acceptance Criteria as defined in Annex H.3.4)		
48	Leakage Test	Leak Rate "A" As Per ISO 5208		
49 50	NDT (MPI / DP / Radiography / UT)	As Per Section 5 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
51 52	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
53 54 55	Painting / External Coating	As Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
56 57	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
58 59	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

60	**Table 11.1 (As Per Valve List)**									
61 62	**S/N**	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
63	1	Elepa Manifold Station	D06	6.0mm	36"	1500#	RTJ FLANGED	SDV1001	ELEP-36"-P-002-D06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
64										

S/N	Datasheet 12: Datasheet for Globe Valve, Carbon Steel, Flanged RTJ, CL 900, 2" to 4"			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	**Vendor's Confirmation**
2	**General Data**			
3	Piping Class	As Per Table 12.1 Below		
4	Valve Type	Tee Type Globe Valve		
5	Valve Size Range	2" to 4"		
6	Valve ASME Class/Rating	*900#*		
7	Valve Port	Regular Bore		
8 9	Valve Service	*Hydrocarbon Gas*		
10	Valve End Connection	*Flanged RTJ*		
11	Corrosion Allowance	**As Per Table 12.1 Below**		
12	Design Temperature (Min/Max)	- 29 / *150* ^{o}C		
13	Design Pressure	Full rating as per ASME B16.34		
14	**Design**			
15	Valve Operation	Handwheel		
16 17	Body Construction	Bolted Bonnet, T-Pattern, Swivel Plug Disc, *Stem Protective Cover & Position Indicator*		
18	Valve Support	Required for valves weighing 250kg or heavier		
19	Lifting Lugs	Required for valves weighing 250kg or heavier		
20	Stem	Rising Stem, OS&Y		
21	Seat / Back Seat / Stem Packing	Renewable		
22	Position Indicator	Required		
23	**Materials**			
24	Body	ASTM A216 Gr. WCB		
25	Cover / Bonnet	ASTM A216 Gr. WCB		
1	05.07.22	Trim 5		
27	Stem	Trim 5		
28	Gland Packing	Filled Graphite		
29	Seat Ring	Trim 5		
30	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.12)		
31	Gland Flange	ASTM A105N		
32 33	Gaskets	SPWD; SS304 Winding Graphite Filled; SS304 Inner Ring; CS Outer Ring		
34 35	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized)		
36	Handwheel	Malleable Steel / Nodular Iron		
37	**Codes and Standards / Testing / Certification**			
38	Valve Design Code	As Per ASME B16.34 / BS 1873 / API STD 623,		
39	Face to Face Dimensions	As Per ASME B16.10		
40	End Connection	As Per ASME B16.5		
41	Fire Safe	As Per API 6FA / BS EN ISO 10497		
42	Hydrostatic & Pneumatic Test	As Per API 598		
43	Low Pressure Gas Seat Test	As Per API 598 at ***6*** barg with Acceptance Criteria in Table 5		
44	Leakage Rate	Leak Rate "B" As Per ISO 5208		
45 46	NDT (MPI / DP / Radiography / UT)	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
47 48	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
49 50 51	Painting / External Coating	SAs Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
52 53	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
54 55	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

56	Table 12.1 (As Per Valve List)								
57 58	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
59	BBOU Manifold Platform	C03	3.0mm	2"	900#	RTJ FLANGED	MV2001	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
60	BBOU Manifold Platform	C03	3.0mm	4"	900#	RTJ FLANGED	MV2002	BBOU-4"-P-004-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01
61	BBOU Manifold Platform	C03	3.0mm	2"	900#	RTJ FLANGED	MV2006	BBOU-6"-P-011-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
62	BBOU Manifold Platform	C03	3.0mm	2"	900#	RTJ FLANGED	MV2004	BBOU-8"-P-009-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
63	BBOU Manifold Platform	C03	3.0mm	2"	900#	RTJ FLANGED	MV2003	BBOU-24"-P-008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
64	BBOU Manifold Platform	C03	3.0mm	2"	900#	RTJ FLANGED	E4R	PIG TRAP	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02
65	Elepa manifold Station	C06	6.0mm	2"	1500#	RTJ FLANGED	MV2001	ELEP-36"-P-002-D06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
66	Elepa manifold Station	C06	6.0mm	4"	900#	RTJ FLANGED	MV2002	ELEP-4"-P-004-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04
67	Elepa manifold Station	C06	6.0mm	4"	900#	RTJ FLANGED	E4R	ELEP-4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
68	Elepa manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	E4R	PIG TRAP. ELEP-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
69	Elepa manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	E4R	ELEP-8"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
70	Elepa manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	E4R	PIG TRAP. ELEP-A-200	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
71	Elepa manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	MV2006	ELEP-6"-P-021-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
72	Elepa manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	MV2004	ELEP-12"-P-019-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
73	Elepa manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	MV2003	ELEP-36"-P-017-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
74	Elepa manifold Station	C06	6.0mm	2"	900#	RTJ FLANGED	E4R	PIG TRAP. ELEP-A-300	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_06
75	Akabel Pigging Station	CO6	6.0mm	4"	900#	RTJ FLANGED	E4R	AKABP -4"-P-016-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
76	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	E4R	PIG TRAP. AKAB-A-100	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
77	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	E4R	AKABP-12"-P-014-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
78	Akabel Pigging Station	CO6	6.0mm	2"	900#	RTJ FLANGED	E4R	PIG TRAP. AKAB-A-100	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05
79	Akabel Pigging Station	CO6	6.0mm	4"	900#	RTJ FLANGED	GV	AKABP-4"-D-002-C06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_05

S/N	Datasheet 13: Datasheet for Globe Valve, Carbon Steel, Flanged RTJ, CL 1500, 2"			
1	**Description**	**Specification Requirement**	**Vendor's Confirmation**	**Vendor's Confirmation**
2	**General Data**			
3	Piping Class	As Per Table 13.1 Below		
4	Valve Type	Tee Type Globe Valve		
5	Valve Size Range	2" to 4"		
6	Valve ASME Class/Rating	*1500#*		
7	Valve Port	Regular Bore		
8 9	Valve Service	*Hydrocarbon Gas*		
10	Valve End Connection	*Flanged RTJ*		
11	Corrosion Allowance	**As Per Table 13.1 Below**		
12	Design Temperature (Min/Max)	- 29 / *150* ^{o}C		
13	Design Pressure	Full rating as per ASME B16.34		
14	**Design**			
15	Valve Operation	Handwheel		
16 17	Body Construction	Bolted Bonnet, T-Pattern, Swivel Plug Disc, *Stem Protective Cover & Position Indicator*		
18	Valve Support	Required for valves weighing 250kg or heavier		
19	Lifting Lugs	Required for valves weighing 250kg or heavier		
20	Stem	Rising Stem, OS&Y		
21	Seat / Back Seat / Stem Packing	Renewable		
22	Position Indicator	Required		
23	**Materials**			
24	Body	ASTM A216 Gr. WCB		
25	Cover / Bonnet	ASTM A216 Gr. WCB		
1	05.07.22	Trim 5		
27	Stem	Trim 5		
28	Gland Packing	Filled Graphite		
29	Seat Ring	Trim 5		
30	Primary / Secondary Seals	Elastomer O-rings, AED (Note 5.12)		
31	Gland Flange	ASTM A105N		
32 33	Gaskets	SPWD; SS304 Winding Graphite Filled; SS304 Inner Ring; CS Outer Ring		
34 35	Bolting	ASTM A193 Gr. B7; A194 Gr. 2H (Hot Dip Zinc Galvanized)		
36	Handwheel	Malleable Steel / Nodular Iron		
37	**Codes and Standards / Testing / Certification**			
38	Valve Design Code	As Per ASME B16.34 / BS 1873 / API STD 623,		
39	Face to Face Dimensions	As Per ASME B16.10		
40	End Connection	As Per ASME B16.5		
41	Fire Safe	As Per API 6FA / BS EN ISO 10497		
42	Hydrostatic & Pneumatic Test	As Per API 598		
43	Low Pressure Gas Seat Test	As Per API 598 at ***6*** barg with Acceptance Criteria in Table 5		
44	Leakage Rate	Leak Rate "B" As Per ISO 5208		
45 46	NDT (MPI / DP / Radiography / UT)	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
47 48	Material Certification	As Per Section 5.7 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
49 50 51	Painting / External Coating	SAs Per Coating / Painting Materials Specification - Linepipes / Pipelines / Field Joints, Structures, and Equipment (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MCM-SPC-010)		
52 53	Marking	As Per Section 4 of Specification for Piping Valves (Doc No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003)		
54 55	Other Applicable Project Spec.	Piping Class Specification. (Doc. No.: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005)		

S/N	Table 13.1 (As Per Valve List)								
57 58	**Location**	**Pipe Class**	**Corrossion Allowance**	**Size**	**Rating**	**End Connection**	**Valve Tag**	**Line Number**	**Reference P&ID Number**
66	Elepa manifold Station	C06	6.0mm	2"	1500#	RTJ FLANGED	MV2001	ELEP-36"-P-002-D06	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04